



School of Future Tech

Case Study Report

on

Airline-Network-Optimisation with route planning

by

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1.

Introduction to the Case Study

Airline Network Optimization System

The Airline Network Optimization System is a smart airline management and ticket booking platform developed using Data Structures and Algorithms (DSA). The project focuses on solving real-world airline operational problems such as route optimization, ticket booking management, seat allocation, dynamic pricing, emergency rerouting, and waitlist handling.

Modern airline systems manage thousands of passengers, routes, and flight schedules daily. Efficient management of these operations requires optimized algorithms and intelligent data handling techniques. This project demonstrates how DSA concepts can be applied practically in airline operations to improve efficiency, reduce operational complexity, and enhance passenger experience.

The system provides two major modules:

1. User Module
Allows passengers to search flights, compare routes and prices, book tickets, select travel classes, and receive smart seat recommendations.
2. Admin Module
Allows airline administrators to manage airports, routes, pricing, emergency rerouting, and crew scheduling.

The project integrates various DSA concepts including:

- Graphs
- Dijkstra Algorithm
- DFS Traversal
- Priority Queue
- Queue
- Stack
- Binary Search
- Greedy Algorithms • BST (Binary Search Tree)

The system is implemented using:

- HTML
- CSS
- JavaScript
- C++

This project demonstrates the practical implementation of DSA concepts in a real-world airline management environment.

Problem Statement / Case Background

2.

(Abstract)

Abstract

Airline companies face several operational challenges such as route optimization, efficient ticket booking, dynamic pricing, seat allocation, and handling emergencies like route blockages or airport closures. Traditional airline management systems may struggle to efficiently process these operations when large datasets and multiple passenger requests are involved.

The Airline Network Optimization System is designed to address these challenges using efficient Data Structures and Algorithms. The system uses weighted graphs to represent airline networks, Dijkstra's Algorithm to determine the cheapest routes, DFS traversal to display all possible routes, and greedy algorithms for dynamic ticket pricing and seat recommendation.

The project also introduces advanced features such as:

- Emergency flight rerouting
- Smart seat recommendation
- Business and economy class booking
- Dynamic pricing based on occupancy
- Waitlist management
- Route and fare comparison

The main objective of the project is to create an intelligent and efficient airline management platform that improves route planning, passenger booking experience, and operational efficiency while demonstrating practical DSA applications.

Problem Statement / Case Study Design

System Design

The system is divided into two primary modules:

3.

1. User Module

The User Module provides passenger-side functionalities.

Features:

- Search flights
- View all routes and prices
- Find cheapest route
- Book tickets
- Select travel class
- Smart seat recommendation
- Ticket cancellation
- Waitlist handling

2. Admin Module

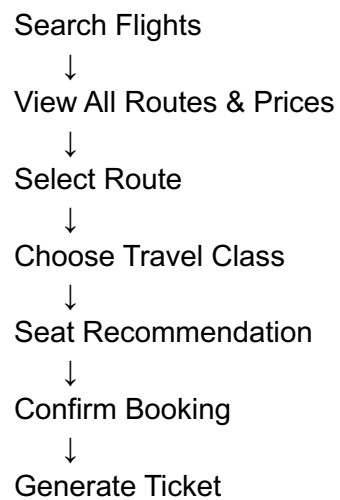
The Admin Module provides airline management functionalities.

Features:

- Add/remove airports
- Add/remove routes
- Manage flight pricing
- Crew scheduling
- Emergency rerouting
- Passenger monitoring
- Demand forecasting

Workflow Design

User Workflow



Admin Workflow



Architecture Design

The project uses a modular architecture:

Frontend:

HTML + CSS + JavaScript

Backend:

C++

Database/File Storage:

Text files / File handling

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

Technologies Used

Technology	Purpose
HTML	Structure of website
CSS	Styling and responsiveness
JavaScript	Frontend interaction
C++	Backend logic and DSA implementation

Data Structures and Algorithms Used

1. Weighted Graph

Used to represent:

- airports as nodes
- flight routes as weighted edges

Purpose:

- route management • network visualization

2. Dijkstra Algorithm

Used for:

- finding cheapest route
- emergency rerouting

Purpose:

- shortest path optimization

3. DFS (Depth First Search)

Used for:

- displaying all possible routes

Purpose: • route

exploration

4. Priority Queue

Used for:

- VIP passenger booking priority Purpose:
- efficient priority handling

5. Queue

Used for:

- waitlist management

Purpose:

- first come first serve passenger promotion

6. Stack

Used for:

- undo cancellation operations Purpose:
- recovery of cancelled bookings

7. Binary Search

Used for:

- fast seat availability search Purpose:
- optimized seat allocation

8. Greedy Algorithm

Used for:

- dynamic pricing
- smart seat recommendation Purpose:
- optimized resource allocation

9. Binary Search Tree (BST)

Used for:

- passenger waitlist storage Purpose:
- fast insertion and retrieval

5. Problem Statement / Case Study Implementation Details Implementation Details

Frontend Implementation

The frontend was developed using:

- HTML
- CSS
- JavaScript

Pages developed:

- Home Page
- Login/Register

- Flight Search Page
- Booking Page
- Admin Dashboard

Backend Implementation

The backend was implemented using C++ modules.

Modules:

- graph.cpp
- dijkstra.cpp
- booking.cpp
- pricing.cpp
- seat.cpp
- rerouting.cpp

Key Functionalities Implemented

1. Flight Route Management

Implemented using weighted graphs.

2. Cheapest Route Finder

Implemented using Dijkstra Algorithm.

Example:

DEL → BLR → HYD

Total Cost: ₹6500

3. Smart Seat Recommendation

Passengers can choose:

- window
- aisle
- near exit

The system recommends the best available seat.

4. Dynamic Pricing

Ticket prices increase based on:

- occupancy
- travel class
- demand

5. Emergency Flight Rerouting

Blocked airports/routes are removed from the graph and alternate routes are generated automatically.

6. Problem Statement / Case Study Results and Conclusion

Results

The Airline Network Optimization System successfully demonstrated the practical application of Data Structures and Algorithms in airline operations.

The system achieved:

- efficient route optimization
- dynamic ticket pricing
- intelligent seat recommendation
- fast booking management
- automatic emergency rerouting
- improved passenger handling

The implementation of Dijkstra Algorithm reduced route search complexity and enabled efficient shortest path calculation. DFS traversal successfully generated multiple route options for passengers. Dynamic pricing and smart seat recommendation improved the realism of the booking system.

The system also demonstrated efficient management of waitlists, cancellations, and travel class handling using Queue, Stack, BST, and Greedy algorithms.

Conclusion

The Airline Network Optimization System is an intelligent and efficient airline management platform developed using modern Data Structures and Algorithms. The project successfully integrates graph algorithms, searching techniques, greedy methods, and queue-based management systems into a real-world airline application.

The project demonstrates how DSA concepts can solve complex operational problems efficiently while improving user experience and system performance.

The system can be further enhanced in the future by adding:

- real-time APIs
- AI-based demand forecasting
- payment gateway integration
- cloud database support
- live flight tracking